

What you say

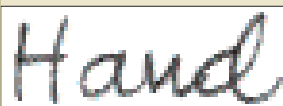
Spoken sentences can be broken into three basic levels: the phoneme level, the word level and the sentence level. A typical SR system analyses speech essentially by identifying phonemes, and looking at how they are strung together. Three steps are required to extract a complete sentence from a speech waveform: identifying the phoneme, the word and the sentence. Most SR systems model these as Hidden Markov Models (HMMs) (See box 'Hidden Markov Models' on Page 100). Alternatives to HMM systems include hybrid systems that use both artificial neural networks (ANNs) and HMMs.

When an HMM is used, the system is constantly making guesses as to what was said. For an English SR system, all the phonemes used in English would be modelled as HMMs. This modelling is done with extensive training from sound dictionaries—for instance, from newspapers and books, spoken by different people with different accents, at different times. After training, the system has a good idea of what each phoneme sounds like. Having heard the letter 'F' thousands of times, spoken at different times by different speakers in reference to different words, the HMM that the system constructs for that sound accounts for variations amongst speakers, and is sufficiently different from the HMM for the sounds of the letter 'S'. Besides, if an SR system is trained by the user, it modifies its phoneme dictionary to reflect the user's speech peculiarities.

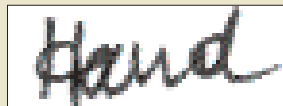
After recognising phonemes, the system needs to put these together into words. This again involves guessing: faced with a sequence of sounds, the system

HR: The Segmentation Problem

One of the most studied problems in HR is that of segmentation: finding the boundaries between letters.



In the text here, segmentation is relatively easy



In contrast, when the word is written this way, segmentation is more difficult



Advanced algorithms make use of heuristics to guess the correct segmentation points

guesses what phonemes must have been uttered to make the sequence. HMMs may again be used in this step. Once the sequence of phonemes has been guessed, it must be constrained into a valid word, for which purpose a dictionary is used. Some systems use HMMs to model the grammaticality of sentences. Such a system would interpret 'he read a book' correctly, instead of as 'he red a book'—because the latter is not a valid sentence, even though the two sound identical.

The written word

Intuitively, it seems that handwriting recognition (HR) is harder than recognising speech. After all, most people pronounce the letter 's' pretty much the same way, but most people don't write 's' the same way, especially doctors! In fact, the immense variation in the way people write is one of the biggest problems in HR. Writing a letter at a time, carefully, with all the right strokes, is not an option we'd like; we want to write in quick scribbles, the way we normally do.

HR technology is driven, in large part, by the emergence of portable devices, and the focus of computing shifting away from the desktop to cell phones, PDAs and tablet

PCs. HR can be notoriously bad on some devices, if we write the way we want, without following the rules; And if we do follow the rules, we know how confining it feels. The lists of rules is long—write on the line, write in mixed upper and lower cases, leave a good gap between words. And when an error occurs, use the keyboard to correct it—which, if it happens too often, defeats the whole point of HR. Systems still need to get to the point where they don't require you to write to their convenience. But how exactly does HR work, and what can we expect in the future?

Recognising handwriting

HR systems vary in the way they detect words, and several methodologies are being explored. In HR, unlike in SR, the basic unit is the word, since there's always a space between words. Some systems work upon the movements the writer made in the process of writing; some work upon the shape of the letters produced. Within the latter category, some systems try to guess a word as a whole, while others try to break apart a word into letters.

Many systems, such as the Tablet PC, use ANNs as part of their recognition engines. ANNs have the advantage of being trainable. To get a feel for how ANNs can be taught simple characters, check out the Java applet at <http://members.aol.com/Trane64/java/JRec.html>. An executable program demonstrating the same idea, developed at Carnegie-Mellon University (CMU), USA, is available at <http://www.andrew.cmu.edu/~jgo/hwrecog.htm>.

The input to the system—say a typical system that uses a neural net, and breaks apart a word into letter segments—is an image of the handwriting to be recognised. This image needs to be pixelised and digitised before it can be used.

The first step the system would perform is the pre-processing of the image—detecting where the handwriting is within the image, removing 'noise' such as stray marks, and possibly the application of algorithms that



Jargon Buster

Phoneme: A phoneme is a basic unit of spoken sound: for example, the sound of the 'f' in 'father', or the sound of the 'ee' in 'sleep'.

Artificial Neural Network (ANN): A layered network of computing units, each of which can perform only a very simple operation. The computational utility of the network comes from the way the units are connected.

Handwriting recognition: The technique by which, for example, words written on a PDA using a stylus, are analysed and converted to text.

Dyslexia: A condition characterised by

difficulty in reading, spelling and so on. This makes it a learning disability. Sometimes, it is accompanied by difficulty in writing as well.

Dysgraphia: A condition characterized by severe difficulty in writing by hand.

Natural Language Processing (NLP): The processing, by a machine, of everyday speech and writing, resulting in the machine getting at the meaning of what was said or written.

Machine learning: The discipline dealing with techniques that enable machines to learn from past experience and data, and possibly predict future data.